

ANALOG CMOS CIRCUIT DESIGN SS26 - PRACTICAL EXERCISE IV

Two-Stage Operational Amplifier RC - Compensation

Tutors

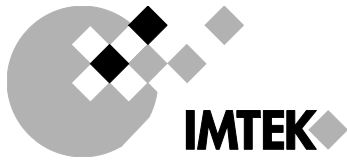
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Analog CMOS Circuit Design SS26

Practical Exercise IV – RC-Compensation

Group

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No review without this standard form!



ANALOG CMOS CIRCUIT DESIGN SS26

PRACTICAL EXERCISE IV

The practical exercise *Analog CMOS Circuit Design* is intended to familiarize you with the design of analog integrated circuits. Based on the example of the two-stage operational amplifier shown in Figure 1, typical design tasks will be performed in the different exercises, i.e.,

- the system theoretical analysis of the circuit
- the transformation of circuit specifications into transistor dimensions
- the simulation and verification of the circuit

To meet the specifications of a circuit, a thorough understanding of its behavior is a must. This exercise is the first step towards the understanding of the two-stage operational amplifier by performing a frequency analysis of the amplifier. To perform this task, the small signal model shown in will be used. **Do not** replace the capacitor C_M (Z_M) with a Miller capacitance, which is connected to V_{SS} !

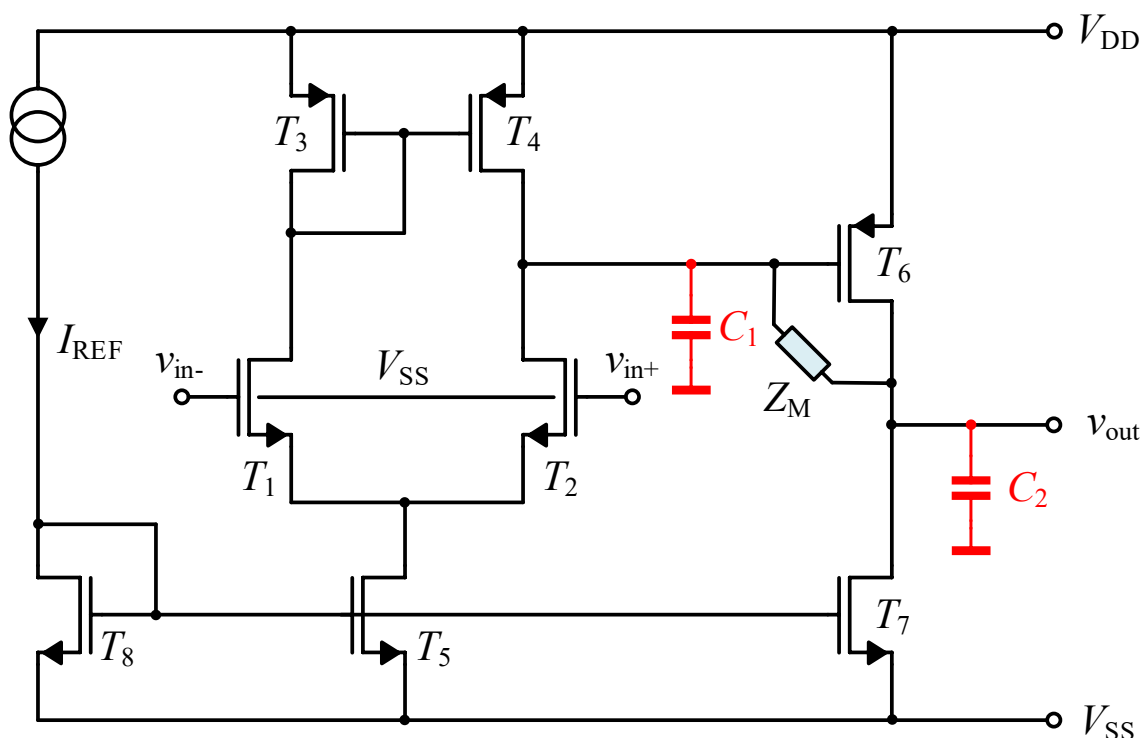


Figure 1: Two-stage Operational Amplifier

This exercise deals with the RC-compensation of the two-stage amplifier as illustrated in Figure 1, which allows either for a larger unity-gain-bandwidth ω_u or a larger phase margin φ_u to be achieved. The reason behind this is that the compensation scheme gives control over the right half-plane zero which is then placed onto the



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second pole in the left half-plane, thus performing a pole-zero compensation. However, it introduces a third pole which must be placed beyond ω_u in order to achieve an appropriate phase margin. On the other hand, the phase margin φ_u may easily be increased by performing a pole-zero compensation while not affecting the unity-gain-bandwidth ω_u .

T1: Small Signal Model

- a) R_1 and R_2 shown in Figure 2 represent the output resistance of the differential amplifier (first stage) and the output resistance of the inverter with current source load (second stage) respectively. Determine their values with respect to T_{1-4} , $T_{6,7}$ in terms of r_{DS} and g_m !
- b) C_1 and C_2 represent the total capacitance from N_1 and N_2 in Figure 2 to ground (N_0) respectively. Determine their values in terms of the parasitic capacitors of the transistors T_{1-4} , $T_{6,7}$ and the load capacitor C_L .

Which parasitic capacitance determines $Z_M = C_M$ assuming a purely capacitive coupled system.

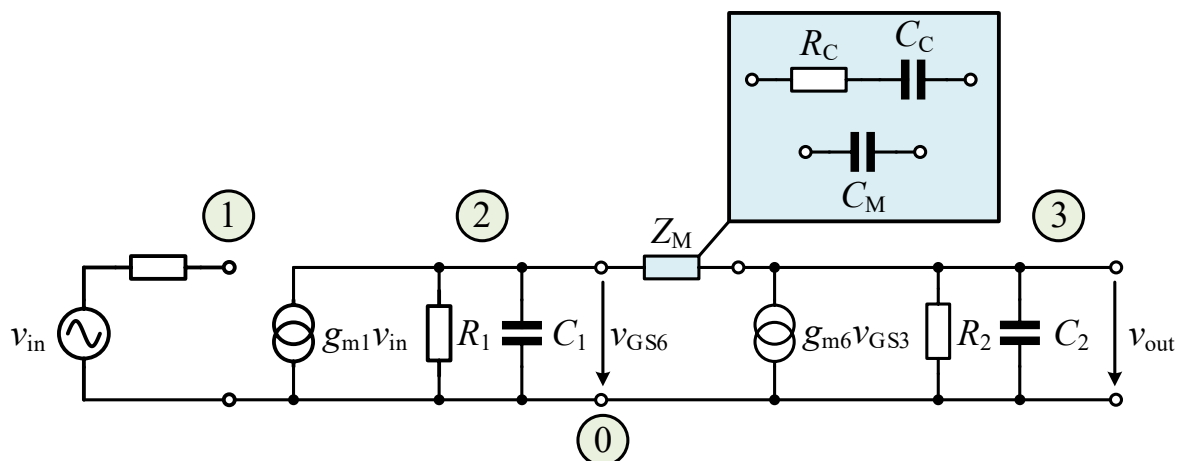


Figure 2: Small signal model of the two-stage amplifier shown in Fig. 1.



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T2: Determination of $v_{\text{out}}/v_{\text{in}}$

In the lecture “Microelectronics”, you have learned that the small signal gain of a circuit is calculated according to

$$\frac{v_{\text{out}}}{v_{\text{in}}} = -g_{\text{m}} r_{\text{out}}$$

using

$$g_{\text{m}} = \left. \frac{\partial i_{\text{out}}}{\partial v_{\text{in}}} \right|_{v_{\text{out}}=0} \quad \text{and} \quad r_{\text{out}} = \left. \frac{\partial v_{\text{out}}}{\partial i_{\text{out}}} \right|_{v_{\text{in}}=0}$$

However, if a circuit becomes more complex, this approach becomes very prone to errors. A more systematic approach to the calculation of the small-signal gain is offered by the approach of the network analysis. This approach allows for the calculation of every node voltage V_1 , V_2 and V_3 in Figure 2 with respect to an arbitrarily chosen reference point (labeled N_0) to be performed. To perform a network analysis, every voltage source must be converted into a current source first!

- a) Convert the voltage source v_{in} and the resistor R_i into a current source (Norton equivalent). Take care of the correct polarity!
- b) Write down Kirchhoff's current law for the nodes labeled 1, 2 and 3, i.e.,

$$\sum_k i_k = 0$$

Incoming currents are positive currents, whereas outgoing currents are negative currents.

- c) Replace every current according to $i = v/Z$, e.g., $i_{C_M} = sC_M(V_2 - V_3)$ at node 2. However, treat Z_M as a generic impedance for now. Replace the corresponding voltages by the corresponding node voltages (e.g., v_{out} by V_3) and derive the system of equations $I = Y \cdot U$ (matrix form).
- d) Solve the linear equation system with respect to V_1 , V_2 and V_3 and derive expressions for V_1/v_{in} , V_2/v_{in} and V_3/v_{in} (Mathematica, Maple, Matlab)



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T3: Determination of zeros and poles

- a) Replace the generic impedance Z_M with a series $R_C C_C$ combination and show that

$$\frac{v_{\text{out}}}{v_{\text{in}}} = \frac{V_3}{V_1} = \frac{b_1 s + b_0}{a_3 s^3 + a_2 s^2 + a_1 s + a_0}. \text{ What are the values of } a_0, a_1, a_2, a_3, b_0, \text{ and } b_1 \text{ in terms of } g_{m1}, R_1, C_1, g_{m6}, C_2 \text{ and } C_C?$$

- b) How many zeros and poles does this transfer function exhibit? What happens if we consider purely miller compensation, i.e. $R_C = 0$?
- c) Determine the zero(s) of the transfer function!
- d) Determine the pole(s) of the transfer function using the following approximation for three widely spaced poles (Mathematica, Maple)

$$\begin{aligned} \left(1 - \frac{s}{p_1}\right) \left(1 - \frac{s}{p_2}\right) \left(1 - \frac{s}{p_3}\right) &= 1 - \left(\frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3}\right)s + \left(\frac{1}{p_1 p_2} + \frac{1}{p_1 p_3} + \frac{1}{p_2 p_3}\right)s^2 - \frac{1}{p_1 p_2 p_3}s^3 \\ &\approx 1 - \frac{1}{p_1}s + \frac{1}{p_1 p_2}s^2 - \frac{1}{p_1 p_2 p_3}s^3 \quad \text{if } |p_1| \ll |p_2| \ll |p_3| \end{aligned}$$

- e) Simplify p_1 , p_2 and p_3 ($C_2 > C_C \gg C_1, R_1 > R_C, R_2 > R_C$).

T3: Compensation & Design

- a) Determine R_C from a pole-zero compensation by placing the zero onto the second pole p_2 .

